

Macroeconomics – Try-Out Exam Answers

AY 2025/2026 | Macroeconomics (MBA Bridging Program)

Prepared from course material

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PART I – Case: Impact of Changes in Electricity Prices

Q 1.1 – Initial Long-Run Equilibrium (Time 0) (0.75 p)

In **long-run equilibrium** all three curves intersect at a single point (P_0, Y^*): actual prices equal expected prices, output equals the natural rate, and unemployment equals its natural rate.

- **LRAS** – vertical at Y^* (determined solely by real factors)
- **SRAS₀** – upward-sloping (sticky wages in the short run)
- **AD₀** – downward-sloping (wealth effect + interest-rate effect)

Time 0 is the single intersection of all three curves in the diagram below.

Q 1.2 – Short-Run Impact of the Breakdown (Time 1) (1.5 p)

A permanent power-station breakdown is an **adverse supply shock**: electricity is an essential production input across the whole economy.

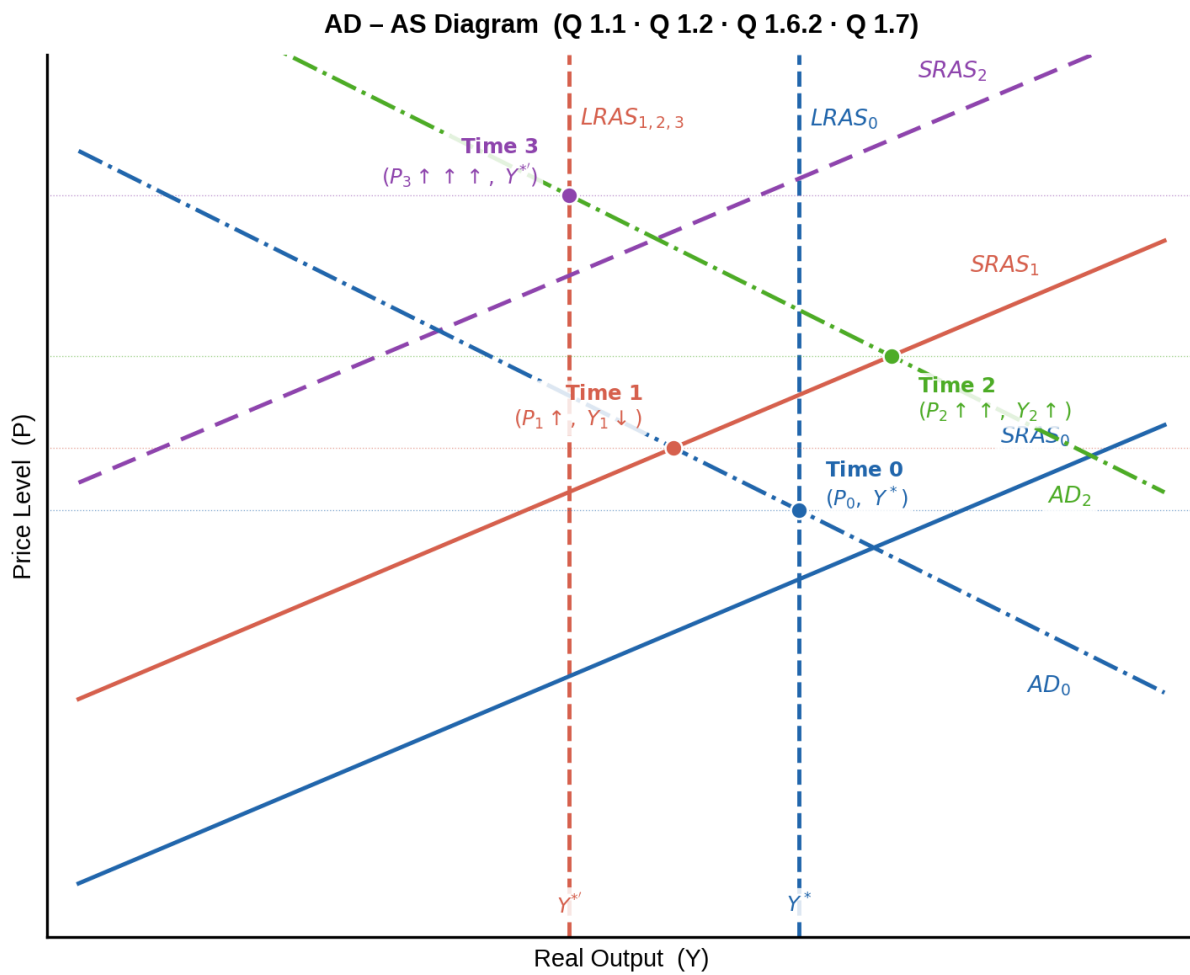
Mechanism:

1. Higher electricity prices raise **production costs** for firms economy-wide.
2. Firms respond by **reducing output** and **raising prices**.
3. The **SRAS curve shifts left / upward** ($SRAS_0 \rightarrow SRAS_1$).
4. Because the breakdown is *permanent*, productive capacity is permanently reduced → the **LRAS also shifts left** ($Y^* \rightarrow Y^{*'}$).
5. AD_0 is unchanged at this stage (no demand-side shock yet).

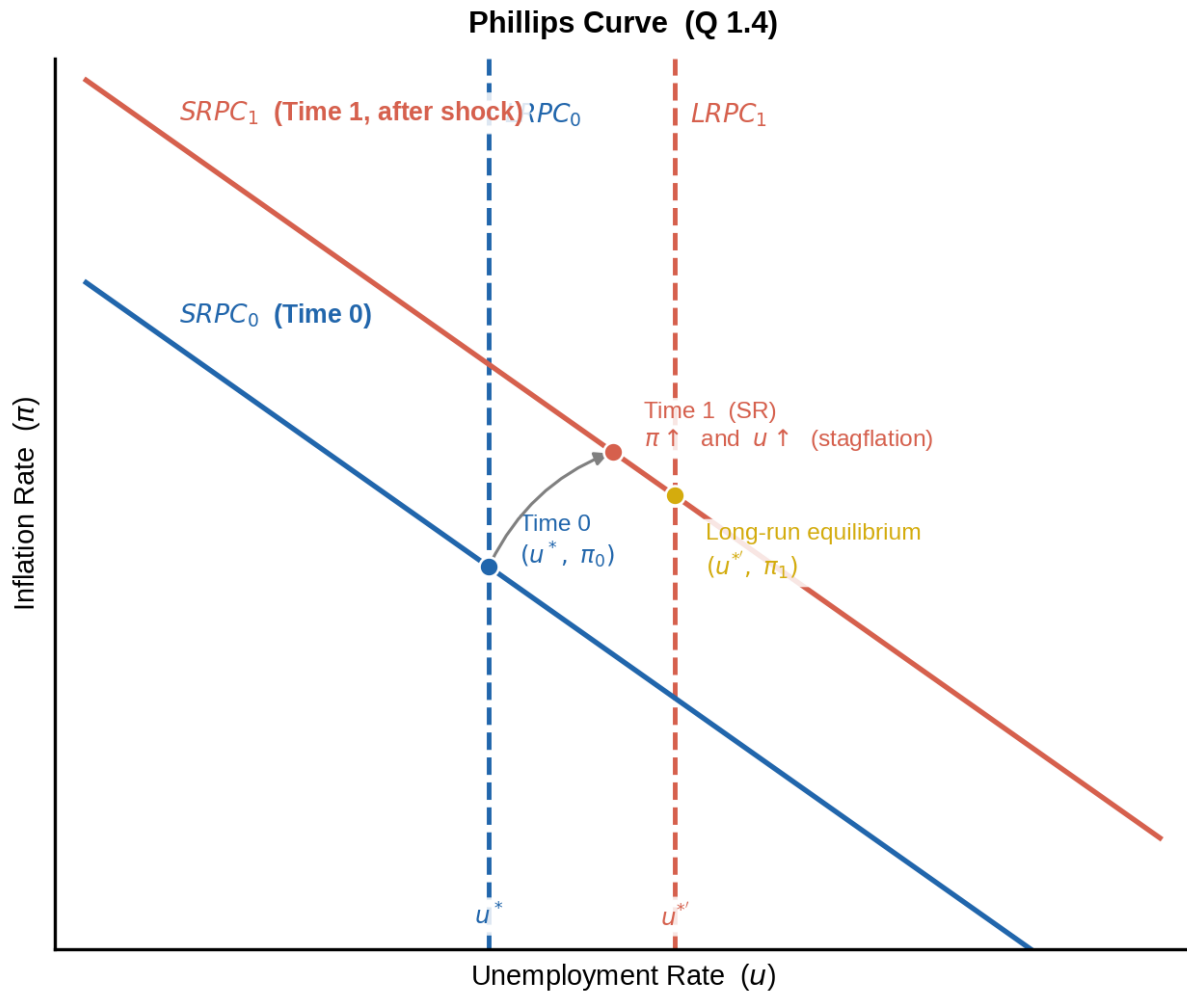
Variable	Short-run direction
Output (Y)	Falls ($Y_1 < Y^*$)
Price level (P)	Rises ($P_1 > P_0$)
Unemployment	Rises

This simultaneous fall in output and rise in prices is called **stagflation**.

AD-AS Diagram (Q 1.1 / Q 1.2 / Q 1.6.2 / Q 1.7)



Q 1.4 – Impact on the Phillips Curve (Time 0 → Time 1) (1.5 p)



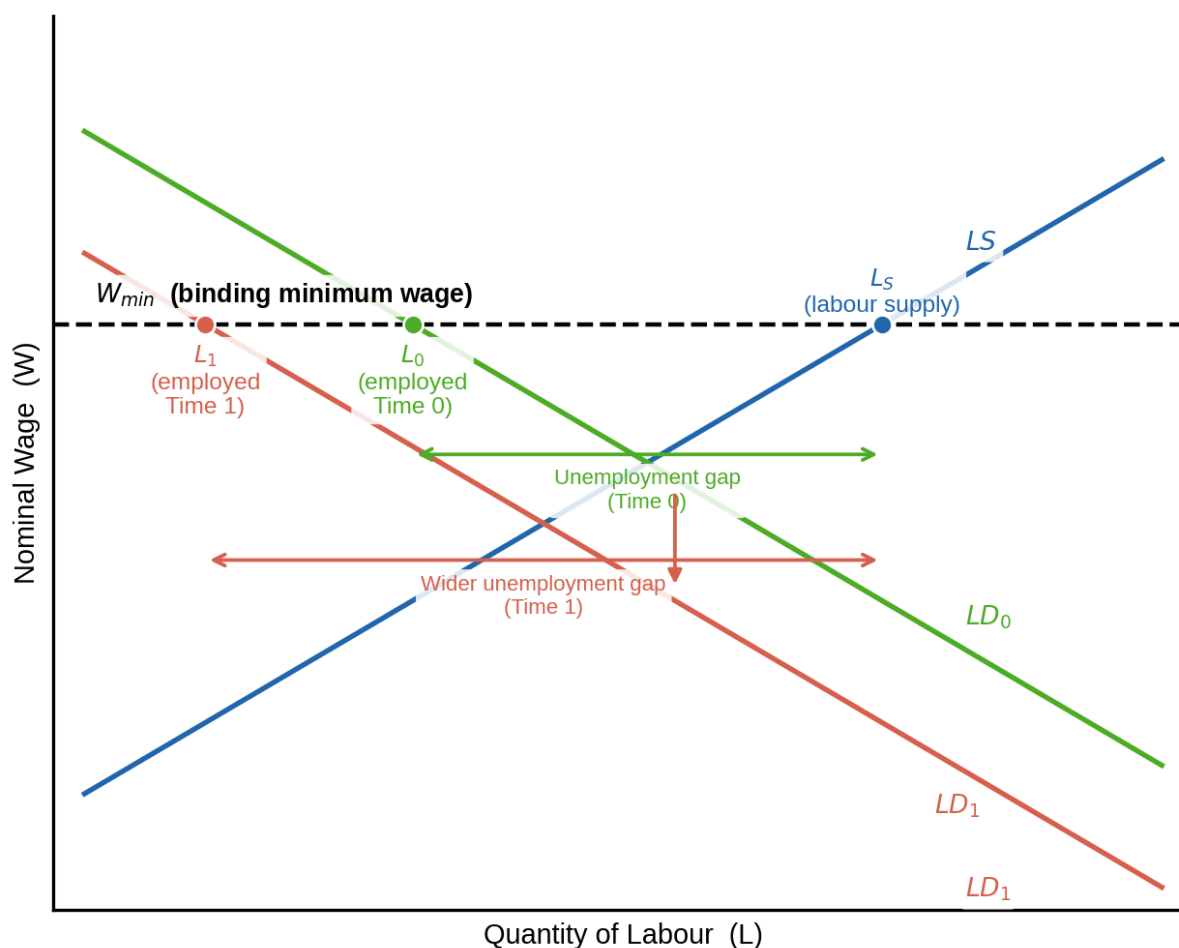
Reading the diagram:

Curve / line	What it shows
$LRPC_0$	Vertical at initial natural unemployment rate u^*
$LRPC_1$	Shifts right to $u^{*'}$ – the permanent breakdown raises the structural natural rate
$SRPC_0$	Initial short-run Phillips curve; Time 0 is on this curve at (u^*, π_0)
$SRPC_1$	Shifts upward – higher input costs push up inflation at every unemployment level
Time 1 (SR)	Economy moves to $SRPC_1$: both inflation and unemployment rise (stagflation)

The supply shock is **unfavourable**: policymakers cannot simultaneously reduce inflation *and* unemployment – the standard trade-off breaks down.

Q 1.5 – Labour Market with Binding Minimum Wage (Time 0 → Time 1) (2 p)

Labour Market – Binding Minimum Wage (Q 1.5)



Initial situation (Time 0):

The minimum wage W_{min} already exceeds the market-clearing wage, so labour supply exceeds demand at W_{min} , creating an existing unemployment gap ($L_S - L_0$).

After the breakdown (Time 1):

- Higher electricity costs reduce firm profitability → firms reduce their **demand for labour** → LD shifts **leftward** ($LD_0 \rightarrow LD_1$).
- Because W_{min} is a legal floor, wages **cannot fall** to clear the market.
- At W_{min} , firms now demand fewer workers: $L_1 < L_0$.

- Labour supply at $W_{\min}$ is unchanged.
- The unemployment gap **widens**: $(L_S - L_1) > (L_S - L_0)$.

Conclusion: With a binding minimum wage the entire adjustment falls on *employment*, not wages – unemployment rises more than in a flexible-wage economy.

Q 1.6 – Government Policy Response

Q 1.6.1 – Type of Policy (0.75 p)

To keep economic activity high before elections the government will implement:

Expansionary Fiscal Policy — an increase in government spending ($\uparrow G$) and / or a cut in taxes ($\downarrow T$).

This directly or indirectly raises aggregate demand, partially offsetting the recession caused by the supply shock.

Q 1.6.2 – Impact on the AD–AS Diagram (Time 2) (0.75 p)

(See the AD–AS diagram above — the green AD_2 curve and the green Time 2 point.)

Expansionary fiscal policy shifts **AD rightward** ($AD_0 \rightarrow AD_2$).

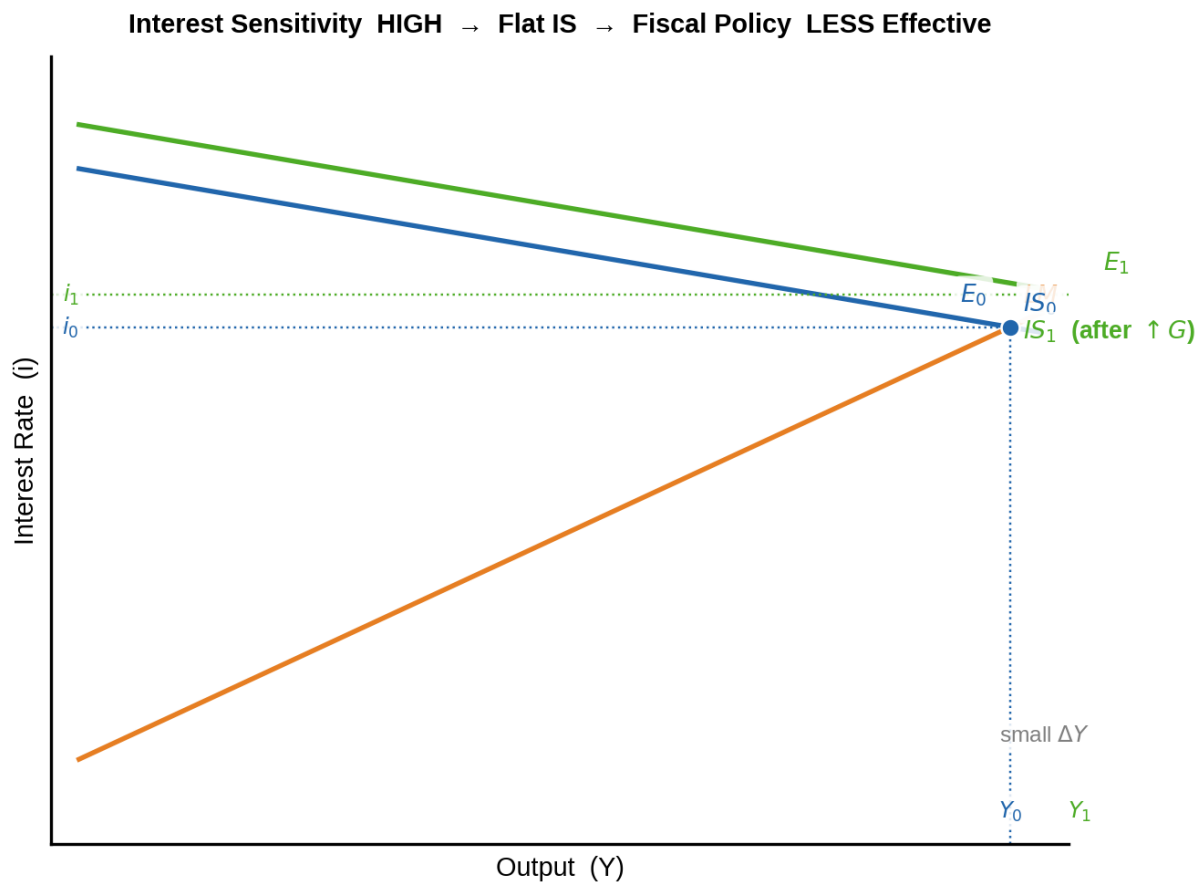
Moving along $SRAS_1$, the new short-run equilibrium (Time 2) shows:

- Output **rises** ($Y_2 > Y_1$, recovering toward Y^*)
- Price level **rises further** ($P_2 > P_1$) — the stimulus adds inflationary pressure on top of the supply shock.

Q 1.6.3 – IS-LM: Fiscal Policy Effectiveness by Interest Sensitivity (2 p)

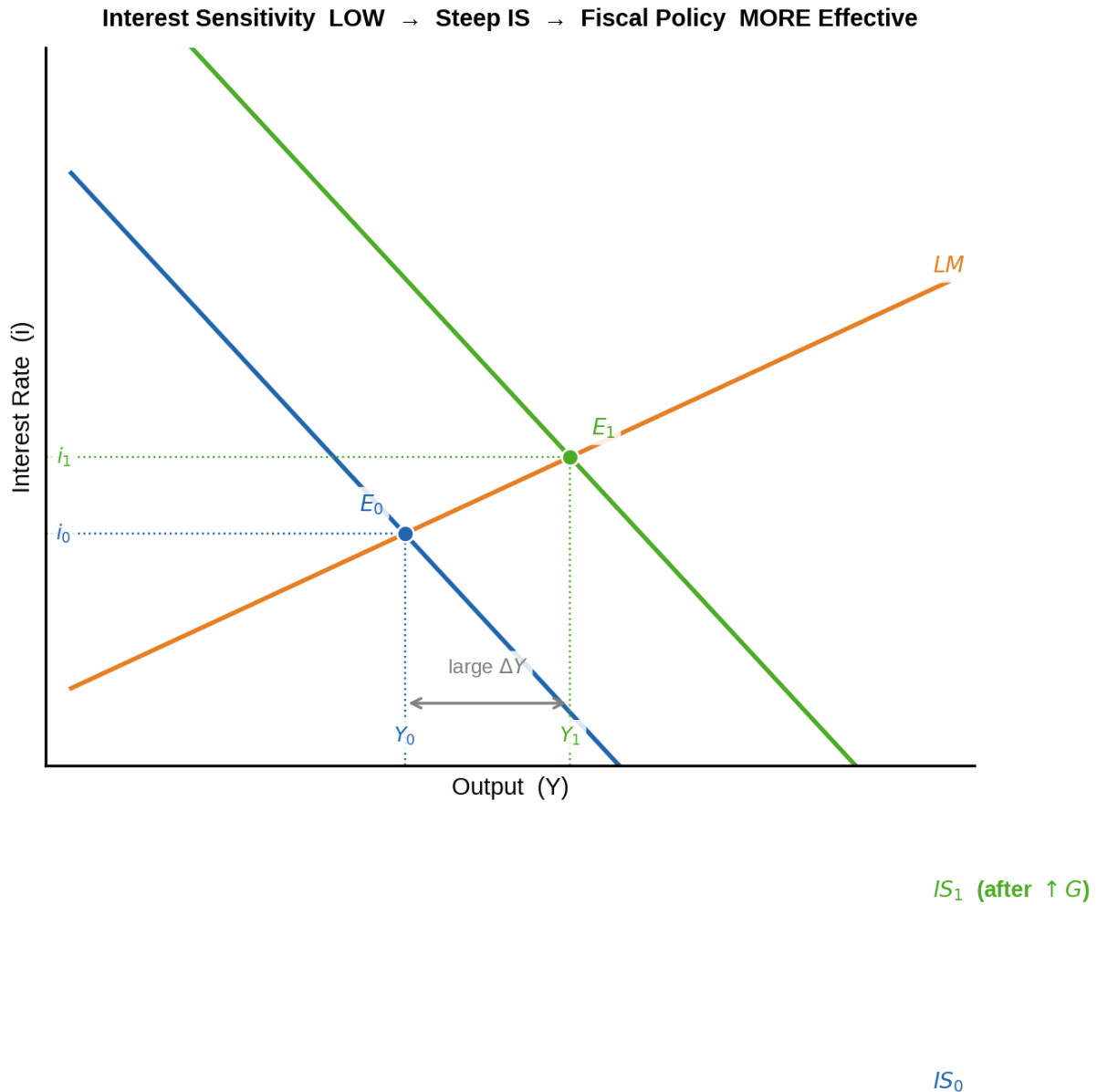
Fiscal policy ($\uparrow G$) shifts the **IS curve rightward**. Higher income raises money demand $\rightarrow$ the interest rate (i) rises $\rightarrow$ private investment is **crowded out**. The degree of crowding-out depends on how sensitive investment is to the interest rate.

When sensitivity is HIGH (flat IS):



A small rise in i causes a large fall in $I \rightarrow$ **large crowding-out** $\rightarrow$ **small ΔY** $\rightarrow$ fiscal policy is **less effective**.

When sensitivity is LOW (steep IS):



A large rise in i causes only a small fall in I → **little crowding-out** → **large ΔY** → fiscal policy is **more effective**.

Interest sensitivity	IS slope	Crowding-out	Fiscal effectiveness
High	Flat	Large	Less effective
Low	Steep	Small	More effective

Q 1.7 – Long-Run Outcome (Time 3) (0.75 p)

(See the AD–AS diagram above — the purple Time 3 point.)

Because the breakdown is **permanent** the long-run adjustment unfolds as follows:

1. At Time 2 output (Y_2) exceeds the new natural rate ($Y^{*'}$) → upward wage and price pressure builds.
2. Workers update their inflation expectations upward → higher wages → higher costs → **SRAS continues shifting left** ($SRAS_1 \rightarrow SRAS_2$) until it intersects the new LRAS at $Y^{*'}$.
3. Long-run equilibrium (Time 3) is where **AD_2 meets the new LRAS**:
 - Output = $Y^{*'}$ (permanently lower than the original Y^*)
 - Price level = P_3 (substantially higher than P_0)

The government stimulus **only raises prices in the long run** — output cannot be permanently kept above the new natural rate.

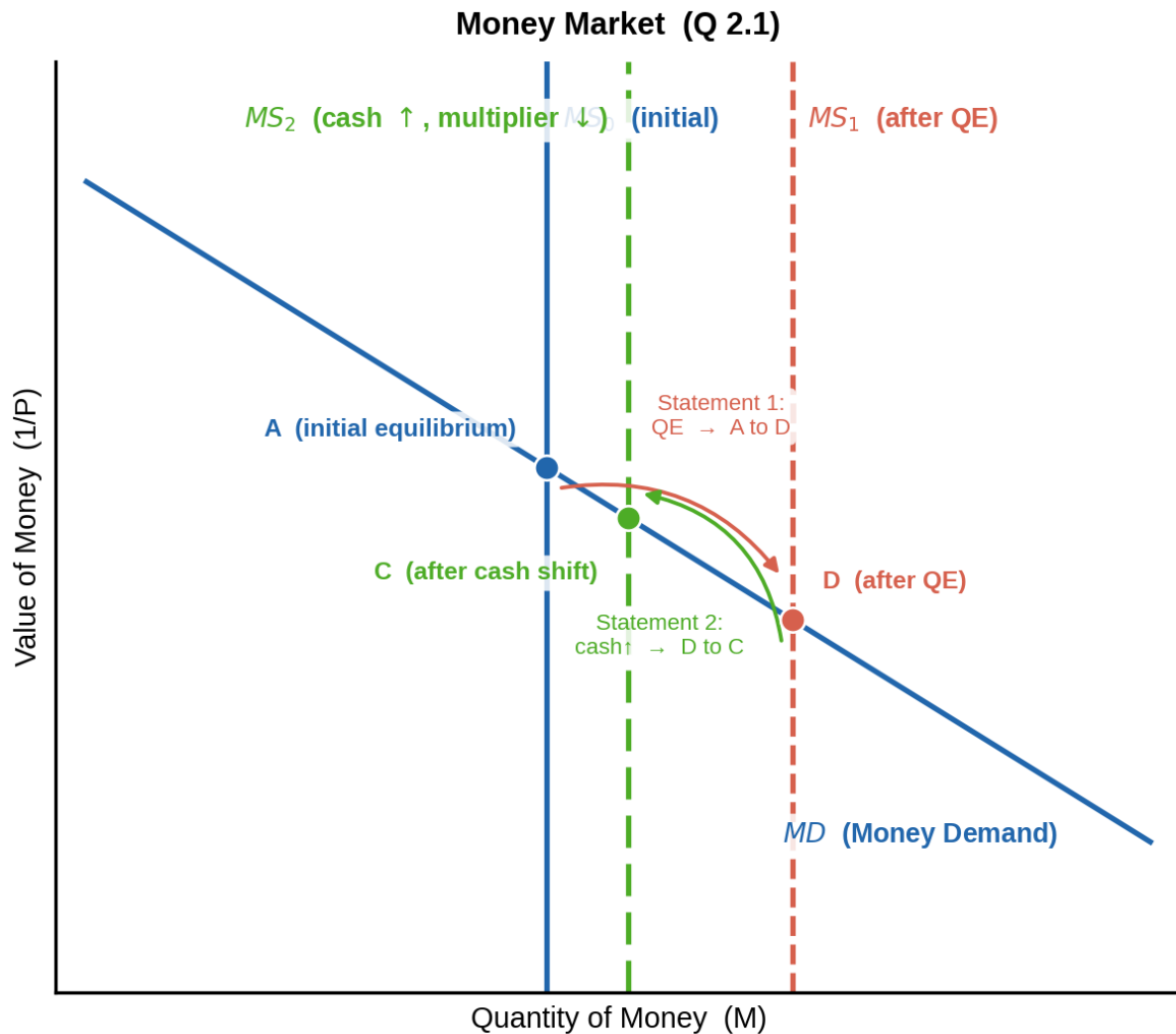
The long-run cost: lower structural output *and* higher inflation.

PART II – Multiple Choice

Answer Sheet

Question	A	B	C	D
2.1	A			
2.2				D
2.3			C	
2.4			C	
2.5			C	

Q 2.1 – Money Market Answer: A — Both statements are correct



Statement 1 – A $\rightarrow$ D via Quantitative Easing: Correct

QE = the central bank buys long-term assets from banks, **increasing the money supply**. The vertical MS curve shifts **rightward** ($MS_0 \rightarrow MS_1$). The new equilibrium D has *more money* and a *lower value of money* (higher price level), which is consistent with the movement A $\rightarrow$ D.

Statement 2 – D $\rightarrow$ C via increased inclination to hold cash over deposits: Correct

When households shift from demand deposits to cash, the **currency-deposit ratio rises** $\rightarrow$ the **money multiplier falls** $\rightarrow$ the effective money supply M1 (**currency + demand deposits**) **contracts** $\rightarrow$ MS shifts **leftward** ($MS_1 \rightarrow MS_2$). The equilibrium moves to *less money* and a *higher value of money*, consistent with the movement D $\rightarrow$ C.

Q 2.2 – Labour Market Statistics Answer: D

	2014	2015
Working-age population (18–65)	10 000 000	11 000 000 (+1M)
Employed	5 000 000	5 000 000 (unchanged)
Unemployed	1 000 000	2 000 000 (doubled)
Labour force	6 000 000	7 000 000
Not in labour force	4 000 000	4 000 000
Unemployment rate	$1/6 = 16.67 \%$	$2/7 = 28.57 \%$
Employment rate	$5/10 = 50.00 \%$	$5/11 = 45.45 \%$
Labour Force Participation	$6/10 = 60.00 \%$	$7/11 = 63.64 \%$
Rate		

- **A** – Unemployment rate doubled? Doubled would be 33.33 %; actual is 28.57 % → **False**
- **B** – Employment rate unchanged? $50.00 \% \neq 45.45 \%$ → **False**
- **C** – Not-in-labour-force decreased? $4 \text{ M} = 4 \text{ M}$ (unchanged) → **False**
- **D** – LFPR increased? $60.00 \% \rightarrow 63.64 \%$ → **True**

Q 2.3 – Before-Tax Nominal Interest Rate Answer: C ($15 \% < x \leq 20 \%$)

Given	Value
After-tax real interest rate	12 %
Inflation rate	2 %
Tax rate	12.5 %

Step 1 — Fisher equation (after-tax):

$$\text{After-tax nominal rate} = 12\% + 2\% = 14\%$$

Step 2 — Apply tax:

$$14\% = x \times (1 - 0.125) = 0.875 x$$

Step 3 — Solve:

$$x = \frac{14\%}{0.875} = 16\%$$

Since $15\% < 16\% \leq 20\%$: **Answer C.**

Q 2.4 – Trading Partner Growth Answer: C — Only Statement 2 is correct

Context: Trading partner's growth raises its income → it imports more from country X → **X's exports rise** → **$NX \uparrow$** → **IS shifts right** → **AD increases.**

Statement 1: "Assume $I = f(i, Y)$. The short-run effect is theoretically ambiguous." — **Incorrect**

With $I = f(i, Y)$ the IS-LM equilibrium still gives a unique, higher output Y :

- $NX \uparrow$ shifts IS right → new IS-LM intersection has higher Y and higher i .
- Higher i lowers I (negative), higher Y raises I (positive via accelerator).
- The effect on *investment alone* is ambiguous, but the effect on **total output (Y)** is unambiguously positive — the IS shift rightward with an upward-sloping LM always produces a higher equilibrium Y .

Statement 1 is **wrong**.

Statement 2: "Price level rises more in the long run without automatic stabilizers." — **Correct**

Automatic stabilizers (progressive taxes, unemployment benefits) dampen the AD increase: as income rises, taxes automatically rise and transfers fall → smaller net AD shift → smaller long-run price increase. Without stabilizers the full multiplier operates → larger AD shift → larger long-run price level increase (LRAS is vertical, so all extra AD shows up as higher P in the long run).

Q 2.5 – GDP Investment Component Answer: C

Recall: **$I = \text{Fixed investment} + \text{Change in inventories}$**

Option	Analysis	Affects I in Eurozone year t ?
A British firm builds factory in London	New factory = investment, but the UK is outside the Eurozone	No
B Belgian buys German car (produced year t)	Both in Eurozone; car purchased by consumer = Consumption (C)	No

Option	Analysis	Affects I in Eurozone year t?
C Dutch company sells carpet produced in year t-1	Carpet entered inventory as +I in t-1 ; selling it from inventory in year t = inventory drawdown → negative I in year t	Yes
D Greek resident buys old house	Existing asset; sale of old assets is not new production → does not enter current GDP	No

Option C is the only transaction that directly affects the Investment component of Eurozone GDP in year t — through a **reduction in inventories** (negative inventory investment).

End of Answers